

NET731 Final Project

Azure Cloud Infrastructure Review

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| Project Overview



Environment Purpose

Designed a multi-tier Azure cloud environment to host a secure, scalable web application. The focus was on high availability and network isolation.



Technical Scope

Implemented an infrastructure featuring virtual networks, compute scale sets, and managed database services with strictly defined security boundaries.

| Architecture Diagram

Infrastructure Blueprint

-  **Hub-and-Spoke:** Isolated workloads via subnetting.
-  **Security:** Layered NSGs and Application Gateway.
-  **Persistence:** Azure SQL with private link connectivity.

 Azure Architecture Diagram

| Key Technical Decisions



VM Sizing

Chose **Standard_B2s** instances. Provides burstable performance for dev/test workloads while keeping compute costs low.



NSG Logic

Implemented a "Deny-All" default rule. Explicit Allow rules only for HTTP/S and SSH from designated jumpbox IPs.



Monitoring

Azure Monitor Logs (Log Analytics). Essential for tracing traffic flows and debugging application connectivity issues.

Security and Monitoring

« → Move ▾ Delete Refresh Give feedback

^ Essentials

Resource group (move) : [myResourceGroup](#) Custom security rules : 0 inbound, 0 outbound

Location : [East US](#) Associated with : 0 subnets, 0 network interface

Subscription (move) : [Contoso Subscription](#)

Subscription ID : [aaaa0a0a-bb1b-cc2c-dd3d-eeeeee4e4e4e](#)

Tags (edit) : [Add tags](#)

Filter by name Port == all Protocol == all Source == all Destination == all Action

Priority ↑↓	Name ↑↓	Port ↑↓	Protocol ↑↓	Source ↑↓	Destination ↑↓
▼ Inbound Security Rules					
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any
65500	DenyAllInBound	Any	Any	Any	Any
▼ Outbound Security Rules					
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork
65001	AllowInternetOutBound	Any	Any	Any	Internet
65500	DenyAllOutBound	Any	Any	Any	Any



Identity Management: RBAC through Entra ID ensures only authorized admins modify resources.



Insight: Azure Bastion for secure management access without public IP exposure.



Alerting: Configured metrics alerts for CPU usage > 80\%\$ and NSG flow logs.

| Cost Management

\$122.45

Total Estimated Spend

Financial Controls

-  **Budgets:** Set soft limit alert at 50% and hard cap at 80%.
-  **Efficiency:** Automatic VM shutdown schedules for non-working hours.
-  **Tagging:** Applied 'Environment' and 'Owner' tags for granular billing analysis.

| Challenges & Solutions

The Problem

Connectivity failure between the Web Subnet and Database Subnet due to overlapping CIDR blocks and NSG misconfigurations.

The Solution

Redesigned VNet addressing and utilized Azure Network Watcher (IP Flow Verify) to identify the specific blocked port.

| Lessons Learned



Governance

Learned that clear resource naming conventions and tagging are as important as the code itself.



Security First

Security must be integrated from the "Architecture Diagram" phase, not added as a feature at the end.



Automation

Developed skills in ARM/Bicep templates for repeatable and error-free infrastructure deployment.

Project Conclusion

The NET731 project successfully demonstrated the implementation of a **highly secure, well-monitored, and cost-efficient** Azure environment. The infrastructure is fully documented and ready for final submission.

"Cloud computing is not just about the technology, it's about the precision of architecture."

| Image Sources



<https://learn.microsoft.com/en-us/azure/architecture/example-scenario/dataplate2e/media/azure-analytics-end-to-end.svg>

Source: learn.microsoft.com



Thumbnail for <https://learn.microsoft.com/en-us/azure/virtual-network/media/manage-network-security-group/network-security-group-details-inline.png>

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